



BUILD — WITH — भारत 2.0

NATIONAL LEVEL HACKATHON



TEAM NAME: Team Paradox

PROBLEM STATEMENT: Smart Digital FIR & Complaint Management System

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PROBLEM STATEMENT

Core Requirement

Develop a secure, accessible and intelligent digital system that simplifies complaint/FIR registration for citizens, while giving police authorities centralized tools for case management, tracking, evidence handling and AI-powered information retrieval from digitized records.

The Problem

Citizens often face difficulties while reporting incidents and interacting with police systems because of:

- Manual and time-consuming complaint registration
- Language barriers between citizens and authorities
- Difficulty in writing detailed complaints
- Lack of convenient voice-based complaint registration
- Limited transparency in complaint / FIR status
- Difficulty managing digital evidence securely
- Delayed identification of emergency situations
- Fragmented police station-level case management



SOLUTION

We propose a centralized Digital FIR & Complaint Management System connecting citizens, an intelligent AI processing layer, police officers and the police station head into one accessible, transparent and secure workflow.



Citizen

Complaint registration, voice input, language selection, evidence upload and status tracking.



Intelligent Processing Layer

Speech-to-text, translation, classification and emergency detection.



Police Officer

Verification, FIR registration, investigation, evidence handling and status updates.



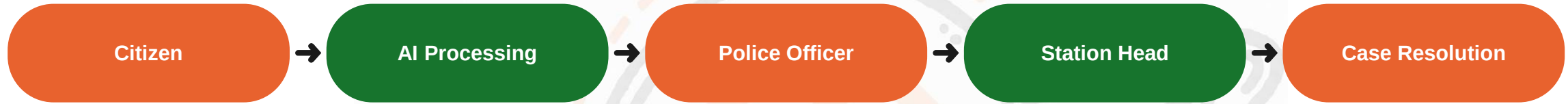
Police Station Head

Centralized database, officer assignment, monitoring and analytics.

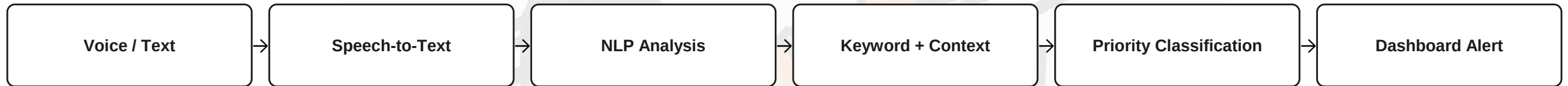


FLOW OF SOLUTION

End-to-End System Flow



Emergency Detection Pipeline



Example

● **Normal** "My bike was stolen yesterday."

● **Critical** "Someone is attacking me right now. I need help."

Voice & language intelligence: Citizen speaks (e.g. Marathi) → Speech-to-Text → Translation → structured complaint with category, object, location, language & priority — original audio is always preserved.



TECH STACK



Frontend

- Android / Web Application
- HTML, CSS, JavaScript / React



Backend

- Java Spring Boot
- or Python FastAPI



Database

- MySQL
- PostgreSQL



AI / ML

- Speech-to-Text & Language Detection
- Machine Translation
- NLP Classification & Priority Detection



Storage

- Secure cloud / object storage for evidence



Security

- HTTPS, Authentication, RBAC
- Encryption & Password Hashing
- Audit Logging, File Integrity Hashing



UNIQUE SELLING PROPOSITION (USP)

01

Complaint Registration

Digital submission of complaints with full incident details.

02

Status Tracking

Track complaint / FIR progress live using a unique ID.

03

Multilingual Support

Multiple languages to reduce communication barriers.

04

Voice Complaint

Record in audio; MP3 → Speech → Text conversion.

05

Emergency Detection

Flags potentially urgent complaints for police attention.

06

Secure Evidence Storage

Photos, videos, audio & documents securely tied to cases.

07

Document Digitization

Scans physical police documents into searchable records.

08

AI Police Assistant

Natural-language Q&A over the authorized case database.

09

Centralized Dashboard

Monitor cases, officers and station-level statistics.



FEASIBILITY & COMPETITORS

Feasibility

- Built on widely available, production-proven technology (Spring Boot / FastAPI, MySQL / PostgreSQL, standard cloud storage).
- Designed to integrate with authorized government / police systems rather than replace them.
- Core AI components (speech-to-text, translation, NLP classification) are established, off-the-shelf capabilities.
- Role-based access (Citizen, Officer, Station Head) maps directly onto existing police station workflows.

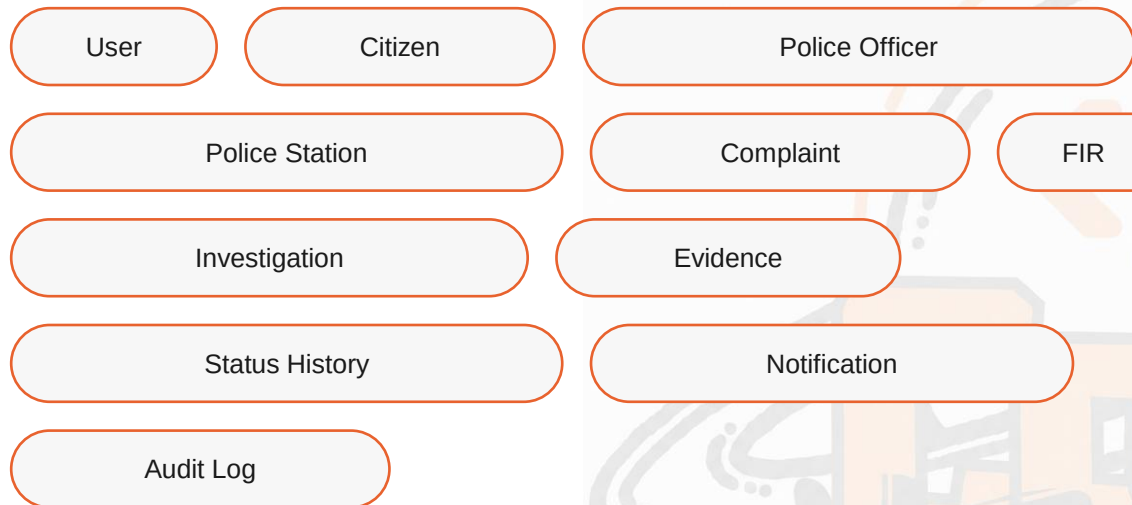
Key Challenges

- Multilingual speech recognition accuracy
- Emergency detection accuracy
- Evidence security
- False emergency alerts
- Legal & procedural compliance
- Scalability



RESEARCH & REFERENCE

Core Database Entities



A full backend data model — not just a UI.

Goals

Accessibility — Make reporting easier for every citizen.

Transparency — Let citizens & authorized personnel track case progress.

Security — Protect sensitive complaints and digital evidence.

GitHub Repository: [Click here to view our GitHub Repository](#)

Demo Link: [Demo link](#)